24 – Glossary & further learning

Definition: The glossary explains technical terms in plain language. “Further learning” shows how you stay on the ball after the training and evaluate new features with a clear head.

> **Remember:** Knowing the terms helps you compare offerings. Evaluate new features first, then use them.

Glossary (approx. 40 terms)

Fundamentals - **AI (artificial intelligence):** umbrella term for systems that solve tasks that usually require thinking. - **Machine learning:** methods in which systems learn from examples instead of being explicitly programmed. - **LLM (large language model):** a large language model; a “text-prediction machine” that predicts the next piece of text from patterns in its training data. - **Model:** the actual “machine” behind the AI; there are small fast variants and large powerful ones. - **Training:** the one-time learning phase of a model; after that it has a knowledge cutoff. - **Token:** the text units that words are broken into; the basis for limits and API billing. - **Context window:** the “working memory” of a chat; it covers the input, the history and the answer. - **Prompt:** your instruction or question to the AI. - **System prompt:** a permanent instruction to the model that the user normally doesn’t see (can be set yourself when using APIs). - **Hallucination:** a made-up but convincingly sounding answer (facts, sources, quotes). - **Multimodal:** the model processes several kinds of input (text, image, audio).

Prompting - **Few-shot:** the prompt contains 1–3 examples of the desired style or format. - **Zero-shot:** no examples, just the instruction. - **Role assignment:** “You are …” – sets the perspective and tone, but doesn’t guarantee correctness. - **Meta-prompting:** the AI improves or drafts prompts. - **Iteration:** improving the result over several rounds (giving feedback, refining). - **Structuring:** separators (e.g. `### Text`) separate the instruction from the content. - **Custom instructions:** permanent personal settings for the service (style, preferences).

Technology & settings - **Temperature:** the control for randomness; low = predictable/matter-of-fact, high = creative/more varied. - **Top-p:** picks only the most likely next words up to a threshold. - **Reasoning / thinking mode:** the model thinks through more steps internally before answering; costs time and tokens. - **Tool use:** the AI calls tools (web search, calculator, code) and uses their results. - **RAG (retrieval-augmented generation):** the AI first searches your documents, then answers based on them. - **Web search (as a tool):** the AI calls up current web pages; it doesn’t replace your source check. - **API:** an interface that lets programs talk to a model directly; billing is usually per token. - **API key:** your personal access key for the API; treat it like a password with a credit card attached. - **Caching:** recurring content is stored in a cache and can be processed more cheaply. - **Batch:** bulk jobs that are processed together (often cheaper).

Risks & safety - **Prompt injection:** hidden instructions in external content that manipulate the AI. - **Jailbreak:** a deliberate attempt to circumvent the AI’s safety rules; risky (terms of service, account suspension). - **Guardrails:** rules and filters for inputs, outputs and actions. - **GDPR (DSGVO):** the European framework for handling personal data. - **Personal data:** anything that can be linked to a person. - **Anonymization:** removing or replacing names, places and identifiers before data

goes into a chat.

Agentic systems - **Agent:** an AI that pursues a goal, plans, uses tools and checks results. - **MCP (Model Context Protocol):** an open standard that connects AI apps to programs and data in a uniform way (“USB-C for AI”). - **Skill:** a reusable package of instructions and templates for a task. - **Subagent:** a specialized AI that the main agent entrusts with a subtask. - **Advisor:** a stronger model that advises on tricky decisions but doesn’t execute anything itself. - **Human-in-the-loop:** a human personally confirms actions with consequences. - **Least privilege:** the agent only gets the access that the task really needs.

Trustworthy sources

- **Provider documentation and help pages:** the first stop for features, limits and prices (they change often).
- **Providers’ data protection pages:** what happens to your inputs, how to switch off training usage.
- **Independent tech media and data protection news:** for news and assessments.
- **Official standards and authorities:** e.g. pages on the GDPR and AI regulation for legal questions.
- Be wary of social media tips: often shortened, outdated or promotional.

The new-feature check

For every new feature or new tool, five questions: 1. **Benefit:** Does it solve a real problem of mine, or am I just curious? 2. **Risk:** What data or access does it expose? What happens if something goes wrong? 3. **Cost:** What does it cost (money, limits, time)? 4. **Data protection:** Where do my inputs end up? 5. **Undo:** Can I undo or switch it off?

Only when four or five answers are “harmless”: try it out – in the test environment first.

Learning plan for the next 30 days

Week	Focus
1	Firmly establish one workflow from topic 23; grow the prompt library to 5 entries
2	Go deeper on a topic of your choice (review the cheatsheet + extra exercises)
3	Data protection check: review the settings of every service you use; review your library
4	Evaluate a new feature with the new-feature check and – if it makes sense – test it; take stock

Mini checklist

- ☐ I can explain the 10 most important glossary terms in my own words.

- ☐ I have noted two trustworthy sources for news.
- ☐ I know the new-feature check and apply it.
- ☐ I have written down my learning plan for the next 30 days.

Exercise (5 min.)

Pick three terms from the glossary that you still feel unsure about and explain each one to a fictional person in a single sentence. Whatever you can't explain in one sentence, look up again in the matching cheatsheet.

Next

- Extra exercises: none (topic 24 is the glossary)
- Test: Test F (Final) and the final test
- After that: capstone (`06_capstone/project_instructions`) – your final project
- Related: all cheatsheets – the glossary links the terms to their topics

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